

Chemotherapy options in elderly and frail patients with metastatic colorectal cancer (MRC FOCUS2): an open-label, randomised factorial trial

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Evidence abstract

BACKGROUND Elderly and frail patients with cancer, although often treated with chemotherapy, are under-represented in clinical trials.

We designed FOCUS2 to investigate reduced-dose chemotherapy options and to seek objective predictors of outcome in frail patients with advanced colorectal cancer.

METHODS We undertook an open, 2 x 2 factorial trial in 61 UK centres for patients with previously untreated advanced colorectal cancer who were considered unfit for full-dose chemotherapy.

After comprehensive health assessment (CHA), patients were randomly assigned by minimisation to: 48-h intravenous fluorouracil with levofolinate (group A); oxaliplatin and fluorouracil (group B); capecitabine (group C); or oxaliplatin and capecitabine (group D).

Treatment allocation was not masked.

Starting doses were 80% of standard doses, with discretionary escalation to full dose after 6 weeks.

The two primary outcome measures were: addition of oxaliplatin ([A vs B] + [C vs D]), assessed with progression-free survival (PFS); and substitution of fluorouracil with capecitabine ([A vs C] + [B vs D]), assessed by change from baseline to 12 weeks in global quality of life (QoL).

Analysis was by intention to treat.

Baseline clinical and CHA data were modelled against outcomes with a novel composite measure, overall treatment utility (OTU).

This study is registered, number ISRCTN21221452.

FINDINGS 459 patients were randomly assigned (115 to each of groups A-C, 114 to group D).

Factorial comparison of addition of oxaliplatin versus no addition suggested some improvement in PFS, but the finding was not significant (median 5.8 months [IQR 3.3-7.5] vs 4.5 months [2.8-6.4]; hazard ratio 0.84, 95% CI 0.69-1.01, $p=0.07$).

Replacement of fluorouracil with capecitabine did not improve global QoL: 69 of 124 (56%) patients receiving fluorouracil reported improvement in global QoL compared with 69 of 123 (56%) receiving capecitabine.

The risk of having any grade 3 or worse toxic effect was not significantly increased with oxaliplatin (83/219 [38%] vs 70/221 [32%]; $p=0.17$), but was higher with capecitabine than with fluorouracil (88/222 [40%] vs 65/218 [30%]; $p=0.03$).

In multivariable analysis, fewer baseline symptoms (odds ratio 1.32, 95% CI 1.14-1.52), less widespread disease (1.51, 1.05-2.19), and use of oxaliplatin (0.57, 0.39-0.82) were predictive of better OTU.

INTERPRETATION FOCUS2 shows that with an appropriate design, including reduced starting doses of chemotherapy, frail and elderly patients can participate in a randomised controlled trial.

On balance, a combination including oxaliplatin was preferable to single-agent fluoropyrimidines, although the primary endpoint of PFS was not met.

Capecitabine did not improve QoL compared with fluorouracil.

Comprehensive baseline assessment holds promise as an objective predictor of treatment

benefit.

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